



Bachelor Thesis

Monetary Policy Transmission in the Euro Area: Low-Cash versus High-Cash Economies

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Date: June 2026

School of Business and Economics

Bachelor

Abstract

The European Central Bank sets a single policy rate for an area in which the share of point-of-sale transactions settled in cash ranges from around twenty per cent in the Netherlands to around sixty per cent in Austria and Italy. Because the standard channels of monetary transmission operate through the banking system, and cash payments bypass that system, this thesis asks whether monetary policy transmits more weakly in economies that rely more heavily on cash.

Using high-frequency monetary policy surprises from the Euro Area Monetary Policy Database, I estimate country-level structural VARs for thirteen euro-area economies and pool the resulting impulse responses in a cross-country regression in which the response at each horizon is interacted with the country's cash share. Higher cash usage is associated with a weaker GDP response to monetary policy. The effect is in the predicted direction at the horizons where output reaches its trough, though not statistically significant in the baseline; it is robust to dropping any single country and sharpens once manufacturing intensity is controlled for. With only thirteen countries the evidence is best read as suggestive.

The price response, measured from the end-of-quarter HICP index, is weaker and less clear-cut: higher-cash countries display an early “price-puzzle” response and a delayed weakening of price transmission at longer horizons that is statistically clearer under a core price index. The evidence is therefore consistent with cash moderating the transmission of monetary policy to output more strongly than to prices, and provides tentative empirical support for the view that a digital euro, by drawing cash transactions into the banking system, could strengthen monetary transmission.

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1 Introduction

Payment behaviour differs markedly across the euro area. According to the European Central Bank's most recent survey of consumer payment attitudes, cash is used in roughly twenty-two per cent of point-of-sale transactions in the Netherlands, against fifty-three per cent in Germany and sixty-two per cent in Austria (European Central Bank, 2024). These differences are economically meaningful, because they correspond to systematically different degrees of contact between households and the banking system.

The distinction matters for monetary policy. A card or transfer payment is settled through a bank: a deposit is debited, a merchant account is credited, and the transaction takes place within the financial system that the central bank influences through its policy rate. A cash payment is settled outside that system; the transaction does not pass through a bank balance sheet and is, in that narrow sense, less directly exposed to policy. This thesis examines whether the prevalence of cash payments affects the strength with which ECB monetary policy is transmitted to the real economy, measured through both the real (output) and nominal (price) responses to a monetary shock.

The mechanism rests on the standard taxonomy of transmission channels (Mishkin, 1995). The interest-rate channel operates because borrowing and lending rates respond to the policy rate; the bank-lending component of the credit channel operates because banks adjust the supply of loans funded by deposits; the balance-sheet component operates because higher rates reduce collateral values and borrower net worth; the asset-price channel operates because the value of financial assets responds to interest rates; and a deposits channel operates because banks pass policy changes onto the spread offered on deposits, so that deposit funding, and hence credit supply, contracts when rates rise.

These channels are not equally dependent on the medium of payment. The credit, deposits and balance-sheet channels work through bank deposits and bank-intermediated credit, and it is precisely the deposit base and the volume of bank-intermediated transactions that a high cash share erodes; the interest-rate and asset-price channels operate through prices and portfolios and are weakened more indirectly, to the extent that cash-intensive households hold fewer rate-sensitive assets and less bank credit. The prediction is therefore not that every channel shuts down, but that the bank-funding-dependent channels weaken first and most where cash is most

prevalent, so that a larger cash share should be associated with a smaller real response to a given policy surprise. The prediction is a tendency rather than an identity: at very high interest rates, for example, the incentive to move out of cash and into interest-bearing deposits rises, which would attenuate the effect.

This reasoning gives the thesis its central research question: *within the euro area, does monetary policy transmit more weakly to output and prices in economies that rely more heavily on cash?* The mechanism yields two testable hypotheses. First, because the bank-dependent channels move real activity, a higher cash share should be associated with a weaker output response to a monetary surprise (H1). Second, to the extent that a weaker real response feeds through to prices, a higher cash share should also be associated with a weaker price response (H2). The two hypotheses are not symmetric: prices respond to a monetary shock more slowly and less completely than output, so H2 is expected to hold, if at all, more weakly and with a longer delay than H1, an asymmetry the results bear out.

The question carries direct policy relevance. The ECB is developing a digital euro, a retail central bank digital currency intended to complement cash (Lane, 2025). If cash usage weakens transmission, replacing part of cash circulation with a bank-connected digital instrument would strengthen the central bank’s influence over the economy, and the magnitude of any such gain would bear on the design choices (holding limits, remuneration, privacy) currently under debate.

A substantial literature documents that monetary policy transmits unevenly across euro-area countries, attributing the heterogeneity to factors such as the share of liquidity-constrained households and differences in financial structure (Almgren, Gallegos, Kramer, & Lima, 2022; Mandler, Scharnagl, & Volz, 2016). The dependence of transmission on financial-system development and inclusion has itself been studied mostly in developing and emerging economies, where cash is more prevalent and financial systems weaker (Mehrotra & Yetman, 2015; Mishra, Montiel, & Spilimbergo, 2010). For the euro area, however, and to my knowledge, no study has examined whether payment structure, among the best-documented features of how households interact with the financial system, accounts for part of this variation, nor connected the question to the digital euro now under development. This thesis addresses that gap, using cash usage in place of the household-level moderators considered in earlier work.

The main result is that the effect operates in the predicted direction for output: countries with higher cash usage exhibit weaker GDP responses. The effect is in the predicted direction

at the horizons where output reaches its trough and is robust to dropping any single country, though it is not statistically significant in the baseline; it sharpens once manufacturing intensity is controlled for. The price response is weaker and more mixed. The remainder of the thesis proceeds as follows. Section 2 reviews the relevant theory and evidence. Section 3 describes the data and Section 4 the methodology. Section 5 reports the baseline results, Section 6 the robustness analysis, Section 7 discusses interpretation and limitations, and Section 8 concludes.

2 Literature Review

2.1 How Monetary Policy Transmits

The benchmark account of monetary transmission is due to Mishkin (1995), who identifies the channels through which a change in the central bank’s policy rate propagates to the real economy. The interest-rate channel is the most direct: a higher policy rate raises the cost of mortgages, business loans and consumer credit, leading households to postpone consumption and firms to defer investment, so that aggregate demand falls. This channel presupposes that agents borrow at rates that respond to policy.

The credit channel concerns the quantity of credit rather than its price. As Mishkin (1995) sets out, a bank-lending component operates through the supply of loans, while a balance-sheet component operates through borrower net worth: higher rates depress collateral values, raise the external-finance premium, and restrict borrowers’ access to credit. Both components amplify the effect of a rate change beyond the interest-rate channel alone. The asset-price channel works through wealth: a higher policy rate makes fixed-income assets more attractive relative to equities and housing, depressing their prices, reducing household wealth and firm collateral, and dampening expenditure. Finally, the deposits channel, identified by Drechsler, Savov, and Schnabl (2017), operates because banks exercise market power over the liquidity services that deposits provide: when the policy rate rises, banks widen the spread between market rates and the rate paid on deposits, deposit funding flows out of the banking system, and bank lending contracts. This channel is central to the argument of this thesis because cash sits at its very root. In the account of Drechsler et al. (2017) it is cash that serves as the household’s outside option to deposits: a higher policy rate raises the cost of holding cash, which is what allows banks to widen deposit spreads in the first place, so it is precisely the deposit base on which the

channel operates that a high cash share erodes.

What these channels share is that they are intermediated by banks, and several of them (the bank-lending, balance-sheet and deposits channels) depend specifically on bank deposits and on transactions being routed through bank balance sheets. This is the foundation of the thesis: a payment instrument that bypasses banks erodes the deposit base and the volume of bank-intermediated activity, and so weakens precisely those deposit-dependent channels, while affecting the interest-rate and asset-price channels only more indirectly. Figure 1 summarises this structure: a contractionary policy surprise propagates through five channels to aggregate demand and ultimately to output and prices, with the three deposit-dependent channels (shaded) attenuated where cash usage is high.

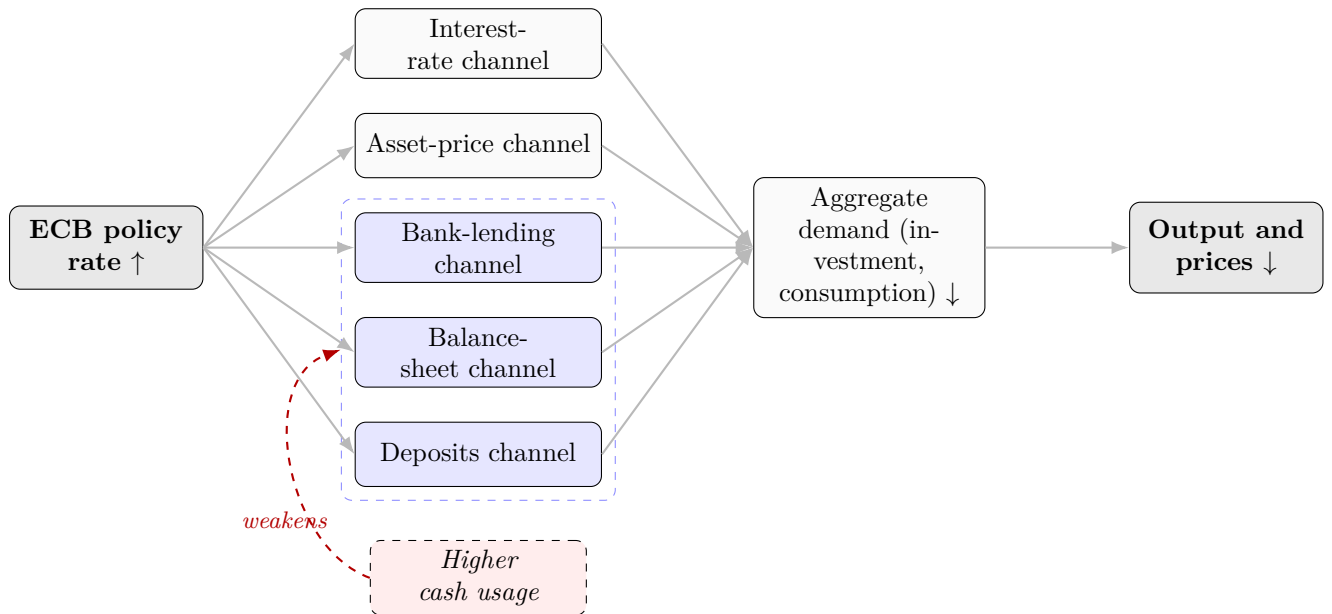


Figure 1: The monetary transmission mechanism. A contractionary ECB surprise transmits through five channels (adapted from Mishkin, 1995) to aggregate demand and, in turn, to output and prices. The three shaded channels (bank-lending, balance-sheet and deposits) operate through bank deposits and bank-intermediated credit; higher cash usage erodes the deposit base on which they depend and so attenuates them (dashed arrow), while the interest-rate and asset-price channels are affected only indirectly.

2.2 Cash and Monetary Policy in Theory

Several theoretical frameworks formalise why the medium of payment should matter. Svensson (1985) develops a general-equilibrium model with a cash-in-advance constraint, in which agents must hold money before purchasing goods. When the constraint binds it drives a wedge between

the marginal utility of wealth and of consumption, and monetary policy operates partly through this wedge; as an economy moves away from cash, the wedge narrows and transmission changes. Lucas and Stokey (1987) distinguish “cash goods”, which require currency, from “credit goods”, which can be purchased through the financial system, and show that monetary policy alters the relative price of the two; payment digitalisation corresponds, in their framework, to a structural shift from cash goods toward credit goods.

Rogoff (2017) makes a more direct argument: because cash offers a zero-return alternative to deposits, agents can escape negative policy rates by hoarding banknotes, which places an effective floor on how negative rates can fall. Although his focus is the zero lower bound, the underlying logic generalises: the larger the share of activity routed through banks rather than cash, the larger the share of the economy that responds when the central bank acts. The financial-inclusion literature reinforces this view. Mehrotra and Yetman (2015) find that greater participation in the formal financial system strengthens transmission, and Mishra et al. (2010) show that weak financial systems impair the standard channels in developing economies. In the euro area, bank-account ownership is near universal, ranging from around ninety-three per cent in Portugal to virtually one hundred per cent in Germany and Austria across the countries studied here (Demirgüç-Kunt, Klapper, Singer, & Ansar, 2022). The relevant margin is therefore not access but use: a consumer who holds an account yet transacts mainly in cash is, for those transactions, effectively outside the banking system. Because account ownership varies so little while cash use ranges from roughly twenty to seventy per cent, the cross-country variation this thesis exploits reflects payment choice rather than differences in access.

2.3 Heterogeneity in the Euro Area

The empirical literature establishes that a common ECB shock produces different responses across member states. Mandler et al. (2016) estimate a large multi-country Bayesian VAR and document substantial variation in output and price responses, attributing part of it to financial structure but not examining payment patterns. The closest methodological precedent is Almgren et al. (2022), who estimate country-level impulse responses using local projections with the EA-MPD instrument and then relate the peak GDP responses to the share of hand-to-mouth households, finding stronger responses where liquidity constraints are more prevalent. Their two-step design (country-level impulse responses followed by a cross-country regression

on a moderator) is the template adopted here, with cash usage in place of the hand-to-mouth share.

The two moderators are related but distinct: a household may be hand-to-mouth yet pay by card, or wealthy yet pay in cash. The hand-to-mouth share captures the sensitivity of spending to income and credit access, whereas cash usage captures how much economic activity is settled outside the banking system. Whether payment structure itself predicts the strength of transmission has not, to my knowledge, been tested. Cash usage is well documented, varies widely across the union, and has a clear theoretical link to the bank-dependent channels, which is the gap this thesis addresses.

2.4 *Cash as a Moderator*

It is worth being precise about the role assigned to cash. Cash usage is treated as a *moderator* of transmission, not as a channel in its own right. A channel lies inside the causal chain from the policy rate to output; cash lies outside it, governing how strongly that chain operates. A payment made by card is intermediated by a bank, contributes to deposit funding, and is connected to the policy rate; the same payment made in cash is not. The larger the share of cash transactions, the smaller the share of activity that monetary policy can influence. Cash does not create a new channel; it attenuates the existing ones. Almgren et al. (2022) apply the same moderator logic with the hand-to-mouth share.

2.5 *The Digital Euro*

The policy motivation is the digital euro, a retail central bank digital currency intended to complement cash. Barrdear and Kumhof (2022) estimate, in a calibrated DSGE model, that issuing CBDC equivalent to thirty per cent of GDP could permanently raise output by around three per cent and improve the central bank's capacity to stabilise the economy. If cash weakens transmission, a digital euro that draws cash transactions into the banking system would extend the central bank's reach, most of all in high-cash economies. The magnitude would depend on design: strict holding limits would displace little cash, whereas an interest-bearing instrument would create a direct channel from policy rates to household incentives. These trade-offs are central to the ongoing design debate (Lane, 2025), and evidence on whether cash actually weakens transmission speaks to it directly.

3 Data

This thesis asks whether a country’s payment structure, the extent to which transactions are settled in cash rather than through banks, affects the strength with which ECB monetary policy reaches its real economy. Testing this requires three components: a common monetary policy shock that can credibly be treated as exogenous; a record of the output and price response of each country to that shock; and a comparable measure of cash use across countries. This section describes each component and the construction choices it involves. All series are brought to a common quarterly frequency and span 1999Q1–2019Q4, with the individual euro-area member state as the unit of analysis.

3.1 Sample Period

The sample begins in 1999Q1, when the euro was introduced and a single monetary policy came into effect; before this date there was no common ECB rate and therefore no shared shock to identify. It ends in 2019Q4, immediately before the COVID-19 pandemic. The contraction in output and prices during 2020 reflected administrative restrictions on economic activity rather than a response to a monetary surprise, and its inclusion would bias the estimated responses.

A further consideration is that the monetary union expanded over the sample period. Each country is exposed to ECB policy only from the date of its euro adoption, so its series begins at the later of 1999Q1 and its accession date. The founding members contribute the full two decades, whereas later entrants contribute shorter samples: Slovakia from 2009, Estonia from 2011, and Latvia and Lithuania from 2014 and 2015 respectively. This staggered accession leaves several countries with short series, a constraint that becomes binding in the methodology, where it determines which countries can be estimated.

3.2 The Policy Shock

Identifying an exogenous monetary policy shock is the central measurement challenge. This thesis uses the Euro Area Monetary Policy Event-Study Database (EA-MPD) of Altavilla, Brugnolini, Gürkaynak, Motto, and Ragusa (2019). Rather than treating a change in the policy rate as a shock (which it generally is not, since the ECB adjusts rates in response to economic conditions), the EA-MPD measures the surprise component of each Governing Council com-

munication from the movement in financial market rates within a narrow window around the announcement, separating the press release from the subsequent press conference. Within such a short window, markets incorporate the policy news and little else of macroeconomic consequence occurs, so the change in interest rates reflects the unanticipated component of the decision.

From the rates recorded in the database, this thesis uses the surprise in the three-month Overnight Index Swap rate (OIS_3M), which captures the revision in expected policy over the coming months and corresponds to the conventional “target” dimension of monetary news rather than to forward guidance or asset purchases. Because the surprises are dated to individual announcements, they are summed within each quarter; quarters without a Governing Council announcement take the value zero. The resulting series is measured in basis points, with positive values denoting an unanticipated tightening.

3.3 Output and Prices

Transmission is defined by the response of two outcomes. Output is real GDP from Eurostat (`namq_10_gdp`), in chain-linked volumes at 2010 prices and seasonally and calendar adjusted. The price level is the all-items Harmonised Index of Consumer Prices from Eurostat (`prc_hicp_midx`), which is constructed to be comparable across member states. The HICP is published monthly, and I use the index level in the final month of each quarter (the end-of-quarter index), so that the quarterly series represents the price level at the end of the quarter and quarterly inflation is a clean change in that level; a quarter is retained only where at least two of its three months are observed. Both GDP and HICP enter the models in natural logarithms, so that the estimated responses can be read as approximate percentage deviations.

3.4 Commodity Prices

In monetary VARs a tightening estimated without a supply-side control frequently appears to *raise* prices, the well-documented “price puzzle”. Theory predicts the opposite: a contractionary shock should depress demand and, with a lag, lower prices, so the puzzle is read not as evidence that tightening is inflationary but as a symptom of imperfect identification. Two interpretations dominate. The conventional one is an omitted-variable problem, in which the central bank tightens in anticipation of cost pressures it observes but the econometrician does not, so the apparent shock partly captures expected inflation. The second is a central-bank information

effect: a high-frequency surprise can confound a pure policy shock with the information the announcement conveys about the economy, so a tightening that markets read as favourable news can raise expected output and prices (Jarociński & Karadi, 2020). The high-frequency surprises used here, with the commodity control introduced next, address the first channel directly; the information effect is harder to purge with this design, and the residual short-horizon puzzle that survives in the price results (Section 5) is consistent with it. To mitigate the omitted-variable channel, the models include the ECB’s euro-denominated, import-weighted index of non-energy commodity prices, covering food, metals and agricultural raw materials, averaged to the quarter and entered in logarithms. The non-energy index is preferred to a single oil price because it more comprehensively captures the input-cost pressures that feed into consumer prices.

3.5 *Cash Usage*

The key explanatory variable is the share of in-person retail payments settled in cash, reported by country in the ECB’s 2024 Study on the Payment Attitudes of Consumers in the Euro Area (SPACE) (European Central Bank, 2024). The measure exhibits substantial cross-country variation, ranging from the low twenties in the Netherlands and Finland to above sixty per cent in several southern and smaller economies; this dispersion is what makes the cross-country comparison feasible. The share is treated as a time-invariant characteristic of each country, on the grounds that payment culture is a slow-moving structural feature and that no historical series of comparable quality is available. Because the measure is drawn from after the sample period, it also reduces the concern that transmission itself shaped payment habits during the sample: payment behaviour recorded in 2024 cannot have been shaped by the transmission dynamics estimated through 2019. The cost of the time-invariance assumption is discussed among the limitations.

3.6 *Robustness Variables*

Three further variables enter only the robustness analysis and not the baseline. A potential confounder is industrial structure. Manufacturing is unusually sensitive to interest rates, so if the manufacturing share is correlated with cash usage across countries, an apparent cash effect could in part reflect differences in industrial composition. Such a correlation is plausible *a priori*: several of the euro-area economies with high cash shares are southern and smaller member states,

whose output tends to be less manufacturing-intensive than that of the core economies, so cash usage and manufacturing intensity may be negatively related across the sample. To address this, each country’s manufacturing share is constructed from Eurostat annual manufacturing gross value added relative to GDP, averaged to a single figure per country over the sample period.

A second potential confounder is trade openness, since monetary policy transmits differently in economies more exposed to foreign trade. Each country’s trade openness is constructed from Eurostat annual exports plus imports of goods and services at current prices (`nama_10_gdp`) relative to GDP, averaged to a single figure per country over the sample period, exactly as the manufacturing share.

The price robustness check additionally uses a core HICP, the Eurostat index excluding energy and unprocessed food (`coicop TOT_X_NRG_FOOD`). It is obtained at monthly frequency and transformed identically to the all-items HICP, as the end-of-quarter index in natural logarithms, and replaces the all-items series only in the price robustness analysis of Section 6.

3.7 Summary of the Data

Table 1 lists the variables used in the analysis, their sources and the transformations applied before estimation.

Variable	Source	Native frequency	Transformation
Monetary policy surprise (OIS.3M)	EA-MPD (Altavilla et al., 2019)	Event / intraday	Summed to quarterly, basis points
Real GDP (B1GQ)	Eurostat <code>namq_10_gdp</code>	Quarterly	Natural log
HICP (all-items)	Eurostat <code>prc_hicp_midx</code>	Monthly	End-of-quarter index, natural log
Non-energy commodity price index	ECB	Monthly	Averaged to quarterly, natural log
Cash share at point of sale	ECB SPACE (European Central Bank, 2024)	Single cross-section	Time-invariant country moderator
Manufacturing share (robustness)	Eurostat (GVA / GDP)	Annual	Sample-average country figure
Trade openness (robustness)	Eurostat <code>nama_10_gdp</code> ((X+M) / GDP)	Annual	Sample-average country figure
Core HICP (robustness)	Eurostat <code>prc_hicp_midx</code> (TOT_X_NRG_FOOD)	Monthly	End-of-quarter index, natural log

Table 1: Variables, sources and transformations.

4 Methodology

The empirical strategy proceeds in two stages. First, the effect of a monetary policy shock is estimated within each country using a structural VAR, yielding an impulse response for output and one for prices. Second, these responses are pooled across countries and related to cash usage. In other words, the responses estimated in the first stage become the data analysed in the second: each country contributes its estimated reaction to a common shock, and the analysis asks whether that reaction is systematically weaker where cash use is higher. This section sets out the identification strategy, the country-level specification, the criteria determining which countries can be estimated, and the cross-country regression that tests the hypothesis.

4.1 Identification

The ECB adjusts rates for several reasons: in anticipation of inflation, in response to output, or in ways that genuinely surprise markets. Only the last constitutes a structural monetary shock, and only it is informative about the effect of monetary policy; a regression of output on the policy rate cannot separate these components. High-frequency identification addresses this. The EA-MPD surprise is the change in market interest rates within an approximately thirty-minute window around an ECB announcement. This window makes the surprise both relevant, as it captures precisely the news that markets had not already priced, and plausibly exogenous, as little else of macroeconomic significance occurs within it. The approach is now standard in euro-area work (Altavilla et al., 2019; Jarociński & Karadi, 2020).

Because the ECB sets a single rate for the entire union, the identified shock is common to all countries; only the responses differ across them. The test of the hypothesis therefore rests on comparing the response paths of economies that rely heavily on cash with those that do not.

4.2 Country-Level VARs

For each country I estimate a four-variable structural VAR in the policy surprise, the log non-energy commodity price index, log real GDP and log HICP:

$$\mathbf{x}_{c,t} = A_1 \mathbf{x}_{c,t-1} + A_2 \mathbf{x}_{c,t-2} + A_3 \mathbf{x}_{c,t-3} + \mathbf{u}_{c,t}, \quad \mathbf{x}_{c,t} = [s_t, lcom_t, lgdp_{c,t}, lhipc_{c,t}]', \quad (1)$$

where s_t is the surprise and c indexes the country. The surprise and the commodity index are common across the union, whereas output and prices are country-specific. The reduced-form residuals (the VAR’s one-step forecast errors, which are mutually correlated) relate to the underlying structural shocks (the economically meaningful, mutually uncorrelated disturbances) through $\mathbf{u}_{c,t} = B\boldsymbol{\varepsilon}_{c,t}$, and the high-frequency surprise, ordered first and used as an internal instrument for its own innovation, identifies the monetary-policy column of B . Concretely, ordering the surprise first makes the monetary structural shock coincide with the surprise innovation, so the corresponding column of B is recovered as the population projection of the reduced-form residuals on the surprise,

$$b_1 = \frac{\mathbb{E}[\mathbf{u}_{c,t} s_t]}{\mathbb{E}[s_t^2]}, \quad (2)$$

which is the impulse vector traced out by the impulse responses. Because the surprise is exogenous by construction, s_t is uncorrelated with the other structural shocks, so this projection is the causal response and no further timing restrictions on output and prices are required. Estimating a separate system per country, rather than a single pooled system, allows the transmission dynamics to differ across countries, which is the heterogeneity of interest; the cost is that cross-border spillovers of the common shock are not modelled (Mandler et al., 2016).

The country responses are estimated by a structural VAR rather than by the local projections used in the template study of Almgren et al. (2022). In country samples as short as forty-four quarters, the parametric structure of a VAR delivers more precisely estimated impulse responses than local projections, which are more robust to misspecification but markedly noisier when the time series is short. Local projections with the surprise as an instrument are therefore reserved as a robustness alternative (Section 7.2), where they would additionally relax the stability requirement the VAR imposes.

Two specification choices warrant justification. First, the systems are estimated in log levels rather than first differences. Real GDP and the HICP are non-stationary in levels, which raises the concern that a level VAR may be near-unstable unless the variables are cointegrated. Estimating in levels is nonetheless standard in the monetary-VAR literature: it yields consistent estimates whether or not the variables are cointegrated (Sims, Stock, & Watson, 1990), allows for cointegrating relationships without pre-testing for them, and preserves the long-run information

that differencing discards. This consistency is, however, an asymptotic property, and in the short post-accession samples of several countries it does not guarantee that the estimated system is stable in finite samples. The implication for the present design is that a country whose system contains a root close to the unit circle will fail the stability condition defined below; this mechanism is behind several of the exclusions in Section 4.3 and is treated there explicitly.

Second, each system is estimated with three lags, with the order reduced to two or one only as a fallback should a three-lag system prove unstable. In practice this fallback is never triggered: every country that enters the baseline is stable at three lags (Table 4), and the four excluded countries are unstable at every lag length (Section 4.3), so three lags are used throughout. Three lags is a compromise between dynamic flexibility and the short post-accession samples of several countries: with four variables, a fourth lag adds four coefficients per equation, which is costly when the shortest usable sample contains only forty-four quarters. The choice is also well-matched to the object of interest: monetary transmission to output is expected to peak within roughly a year, and three quarters of dynamics suffice to generate the hump-shaped impulse responses of that length, whereas additional lags in these short samples tend to fit noise and destabilise the estimated responses. The lag order is fixed at three on these grounds rather than selected by an information criterion, a choice consistent with the parsimonious orders such criteria typically favour for quarterly macroeconomic VARs.

Identification within the VAR follows the logic above. With the system identified, the impulse responses are computed to twenty quarters and expressed in per cent, giving each country a path for output and a path for prices following an unanticipated tightening. The impulse response is the object of interest because it traces, period by period, how a single policy surprise propagates to output and prices; the depth of that response measures the strength with which monetary policy reaches the real economy, which is the quantity the hypothesis compares across high- and low-cash countries.

A VAR is *stable* when every eigenvalue of its companion matrix lies strictly inside the unit circle, that is when its spectral radius satisfies

$$\rho(\mathbf{C}) = \max_i |\lambda_i| < 1, \quad (3)$$

which is necessary for the system to admit a convergent moving-average representation and

hence for the impulse responses, built from the companion matrix raised to successive powers, to remain finite. This condition is also the practical criterion that determines which countries can be estimated: when a country's fitted companion matrix has a spectral radius of one or more, its impulse responses do not converge and the country is excluded from the analysis (Section 4.3). For the cross-country comparison, each path is summarised by its peak response, defined as the trough (the most negative value the response reaches) within the first twelve quarters; the horizon is capped at twelve on the grounds that a monetary shock should have been transmitted within three years and that extending the search introduces late-horizon noise.

4.3 Sample Selection

The staggered accession of the monetary union determines which members can be estimated. Because the excluded countries are not a random sample, it is important to be explicit about which are dropped and why. Of the nineteen members, thirteen yield a usable baseline estimate. The six exclusions arise for two distinct reasons.

Insufficient sample length. A country is excluded if it has fewer than thirty quarters common to all four series. Because each country's sample begins at its euro accession (Section 3), the most recent entrants fall below this threshold: Latvia and Lithuania, in the union only since 2014 and 2015, provide roughly two dozen and fewer than twenty quarters respectively. A four-variable VAR with three lags estimates twelve coefficients and a constant per equation, which cannot be done credibly on so few observations. This criterion is fixed in advance, depends only on the length of the series, and is unrelated to the hypothesis.

No stable estimate. For Estonia, Ireland, Malta and Slovenia, no stable estimate is obtained at any of the three lag lengths. If the spectral radius reaches one, the moving-average sum diverges and the impulse response is not a well-defined object: it grows without bound rather than settling back, so there is simply no finite response to read off and carry into the second stage. These countries are therefore excluded on a mathematical ground, not a discretionary one. As noted above, this instability is closely related to the use of non-stationary variables in levels.

This second exclusion is a form of sample selection, since it conditions on a property of the

fitted model, and it is correlated with country characteristics: the affected countries tend to be small and volatile, and include cases that run counter to the hypothesis. A baseline conditioned on stability may therefore appear tidier than the underlying data. This is a genuine limitation of the design, acknowledged again among the limitations in Section 7.2. The second stage mitigates it only to the extent that it uses every surviving country and every horizon.

Table 2 lists all nineteen members with their cash share, euro-adoption date, the length of the resulting sample, and whether they enter the baseline. The thirteen survivors span almost the full range of cash usage, from twenty-two per cent in the Netherlands to sixty-two per cent in Austria, which is the cross-country variation the second stage exploits.

Country	Cash share (%)	Euro adoption	Quarters	Status
Netherlands	22	1999	84	Baseline
Finland	27	1999	84	Baseline
Luxembourg	37	1999	84	Baseline
Belgium	39	1999	84	Baseline
Estonia	39	2011	36	Excluded (unstable)
France	43	1999	84	Baseline
Latvia	45	2014	24	Excluded (short sample)
Cyprus	48	2008	48	Baseline
Ireland	49	1999	84	Excluded (unstable)
Germany	53	1999	84	Baseline
Greece	54	2001	76	Baseline
Lithuania	54	2015	20	Excluded (short sample)
Portugal	54	1999	84	Baseline
Slovakia	57	2009	44	Baseline
Spain	57	1999	84	Baseline
Italy	61	1999	84	Baseline
Austria	62	1999	84	Baseline
Slovenia	64	2007	52	Excluded (unstable)
Malta	67	2008	48	Excluded (unstable)

Table 2: The nineteen euro-area members, ordered by cash share. “Quarters” is the number of quarters from euro adoption to 2019Q4. “Status” records whether the country enters the baseline or is excluded for a sample shorter than thirty quarters or for the absence of a stable VAR at any lag length (Section 4.3). Cash shares are from the ECB SPACE survey (European Central Bank, 2024).

4.4 The Cross-Country Regression

For each surviving country, a full response path is available for output and for prices. The simplest cross-country test would reduce each country to its peak response and regress this on the cash share, in the spirit of the cross-country literature (Almgren et al., 2022). With only thirteen countries, however, such a regression has limited power and is sensitive to individual

observations, and it discards most of the information in the response paths. The analysis therefore uses the entire path. The impulse responses are pooled into a single regression in which the unit of observation is a country–horizon pair (c, h) , and the response is explained by a full set of horizon indicators together with those indicators interacted with the country’s cash share:

$$y_{c,h} = \sum_h D_h \theta_h + \sum_h (D_h \times \text{cash}_c) \gamma_h + \varepsilon_{c,h}. \quad (4)$$

The horizon dummies θ_h trace out the average response path across countries. The coefficients of interest are the interactions γ_h , each measuring the effect of a one-percentage-point increase in the cash share on the response at horizon h . The hypothesis predicts the sign of these coefficients for both outcomes. For output, γ_h should be positive, since a higher cash share should make the contraction shallower. For prices, γ_h should likewise be positive: a contractionary shock lowers prices, so weaker transmission in higher-cash countries implies a smaller price decline, a less negative response, at the relevant horizons.

Standard errors are clustered by country with a small-sample correction, and the reported bands are ninety-per-cent intervals from a t -distribution with degrees of freedom equal to the number of countries minus one, in line with the cross-country impulse-response regression approach of Almgren et al. (2022). Specifying the test in this way indicates not only whether cash matters but at which horizons in the transmission process its effect appears.

5 Results

5.1 A Pilot: Germany and the Netherlands

The analysis begins with a detailed comparison of Germany (53% cash) and the Netherlands (22% cash) estimated in a single combined VAR. The two are a natural starting pair: they span a wide gap in cash use while both being large, founding euro-area economies with the full two-decade sample, which makes them the most directly comparable high-cash and low-cash economies available, and the cleanest two-country contrast the data allow. The purpose of this pilot is twofold: to establish a specification that resolves the price puzzle and yields stable, interpretable impulse responses, and to assess how much can and cannot be learned from comparing just two economies. Methodologically it is instructive. Early specifications in

log differences produce unstable results, with the GDP comparison reversing across lag lengths; adding oil as a supply-side control resolves the price puzzle for Germany but not the Netherlands; switching to log levels stabilises the comparison; and replacing oil with the broader ECB non-energy commodity index at three lags resolves the price puzzle for both. These choices carry directly into the full specification.

Substantively, however, the pilot is inconclusive, and deliberately so. In the preferred specification Germany responds slightly more strongly than the Netherlands, the opposite of the hypothesis, but the confidence bands overlap (Figure 2) and the two economies differ along too many dimensions (size, openness, industrial structure) for a two-country contrast to identify the role of cash. What the pilot cannot do is precisely what motivates the cross-country design: isolating cash requires variation across many economies and a moderator that varies systematically with it, which is the purpose of the all-country analysis that follows.

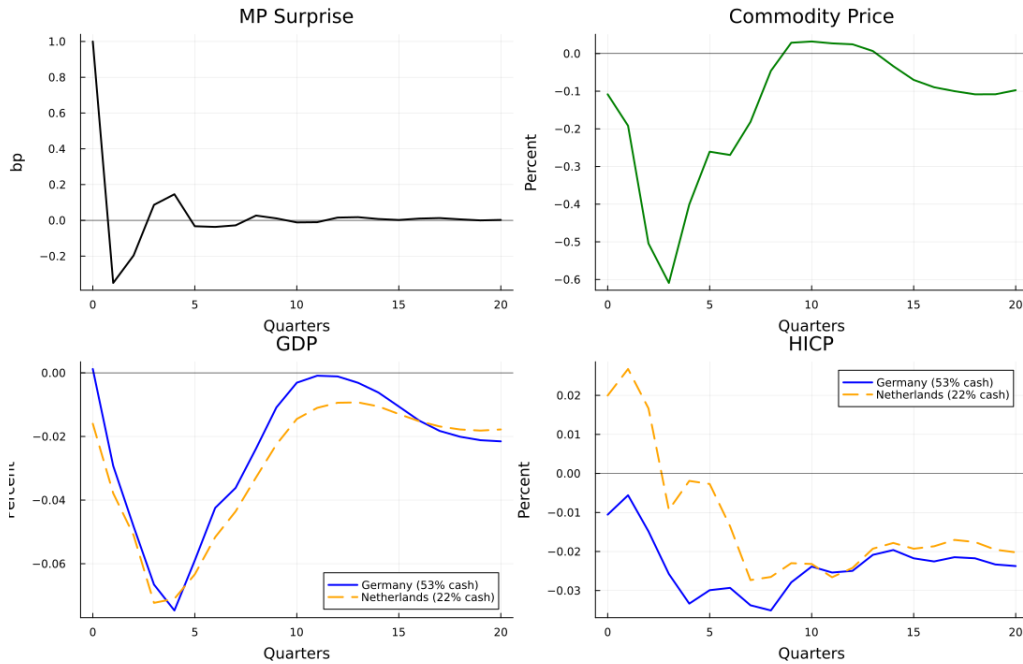


Figure 2: Pilot comparison of Germany (53% cash) and the Netherlands (22% cash) in a combined VAR estimated in log levels with the ECB non-energy commodity index and three lags: impulse responses to a one-basis-point contractionary monetary policy surprise. The output responses are nearly identical (Germany a fraction deeper), illustrating why a two-country contrast cannot identify the role of cash.

5.2 All-Country Estimates

Of the nineteen members, thirteen yield stable VAR estimates with a sufficient sample: the ten founding members other than Ireland, together with Greece, Cyprus and Slovakia. The remaining six are excluded either for short samples, with fewer than thirty common quarters (Latvia, Lithuania), or because no stable estimate emerges at any lag length (Estonia, Ireland, Malta, Slovenia), as detailed in Section 4.3. Twelve of the thirteen surviving countries show GDP declining after a contractionary shock, broadly confirming that the shock is properly identified; Greece is the exception, with a small positive response over the first twelve quarters. The individual country impulse responses are shown in Figure 3.

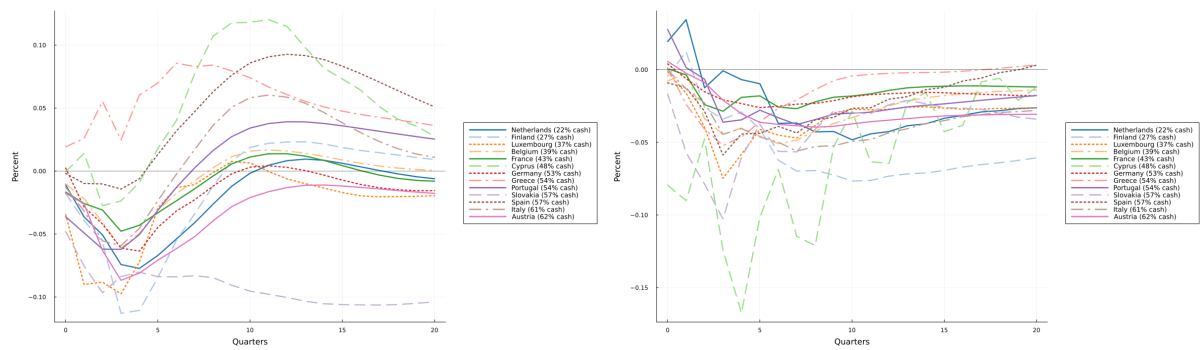


Figure 3: Country-level impulse responses of GDP (left) and the HICP (right) to a one-basis-point contractionary monetary policy surprise, for all thirteen surviving countries. The horizontal axis is the horizon in quarters after the shock and the vertical axis the response in percent; each line is one country, labelled in the legend with its cash share. Lines are distinguished by colour and style, and the figure is intended to convey the broadly shared contraction across countries rather than the path of any individual one.

5.3 The Cash Effect on Output

In a pooled specification constraining the cash effect to a single coefficient across all horizons (equation (4) with $\gamma_h = \gamma$ for all h), the estimate is positive but not statistically significant ($\gamma = 0.00042$, $p = 0.56$); averaging across twenty-one horizons dilutes the signal. Allowing the effect to vary by horizon concentrates it where theory expects it. As Figure 4 and Table 3 show, γ_h is positive across quarters three to five, the horizons at which the average output response troughs, with estimates of 0.00111, 0.00130 and 0.00105. These are economically meaningful and correctly signed, but not statistically significant ($p = 0.151$, 0.133 and 0.234): with only thirteen countries the cash share is imprecisely estimated. At impact the effect is small, and

beyond quarter six it fades.

Because a coefficient is estimated at each of twenty-one horizons, the individual p -values invite a multiple-comparisons concern, so the primary inferential statement is a joint test across the trough window. Testing whether the cash interaction is zero *on average* over quarters three to five, with the same country-clustered covariance, the average effect is positive but not significant ($\bar{\gamma} = 0.00115$, $t_{12} = 1.50$, $p = 0.159$). The average is preferred to a multi-coefficient Wald test because, with only thirteen clusters, the joint cluster-robust covariance is poorly conditioned and the multi-degree-of-freedom test unreliable. The window is not chosen for its significance: quarters three to five are the pre-specified trough of the average output response, where transmission is expected to peak.

The data thus support a consistent, correctly signed association, concentrated at the peak-transmission horizons and spread across three contiguous quarters rather than an isolated one, and robust to dropping any single country (Section 6), but short of conventional significance in the baseline. This is the pattern expected if cash attenuates the real response, and is reported as suggestive rather than conclusive.

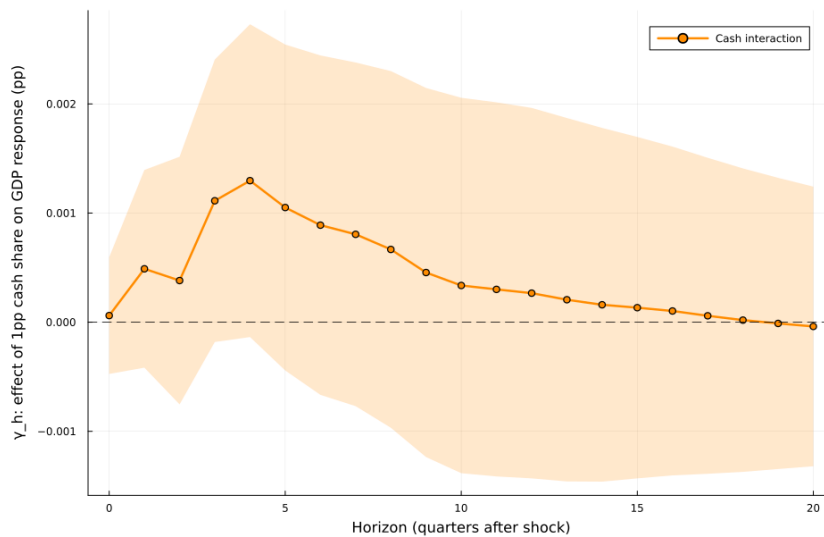


Figure 4: Horizon-varying cash interaction γ_h for the GDP response, with 90% bands (standard errors clustered by country). Positive values indicate that higher cash usage is associated with a shallower output contraction.

Horizon	Baseline γ_h	p (baseline)	With manufacturing	p (with manuf.)
q3	0.00111	0.151	0.00142	0.018
q4	0.00130	0.133	0.00172	0.009
q5	0.00105	0.234	0.00153	0.023

Table 3: Cash interaction γ_h for the GDP response at the trough horizons, baseline and with the manufacturing control. Positive values indicate that higher cash usage is associated with a shallower contraction. Standard errors clustered by country; $G = 13$.

5.4 The Cash Effect on Prices

The price results are weaker and more mixed than those for output. Under the end-of-quarter HICP, the cash interaction is significantly *negative* at the shortest horizon (quarter one), a stronger price response in higher-cash countries at impact; this is an early price-puzzle-like pattern rather than support for the hypothesis. At longer horizons the sign reverses, γ_h turning positive around quarters ten to fourteen, consistent with delayed weaker price transmission, though under the all-items index this long-horizon effect is not statistically significant (Figure 5). The price evidence is thus weaker, less precise and less robust than the output evidence: cash appears to moderate transmission to output at the trough, but to prices only weakly and with a long lag. Adding the manufacturing control of Section 6 leaves the profile broadly similar, retaining the early negative response and the delayed reversal (Figure 5, right panel).

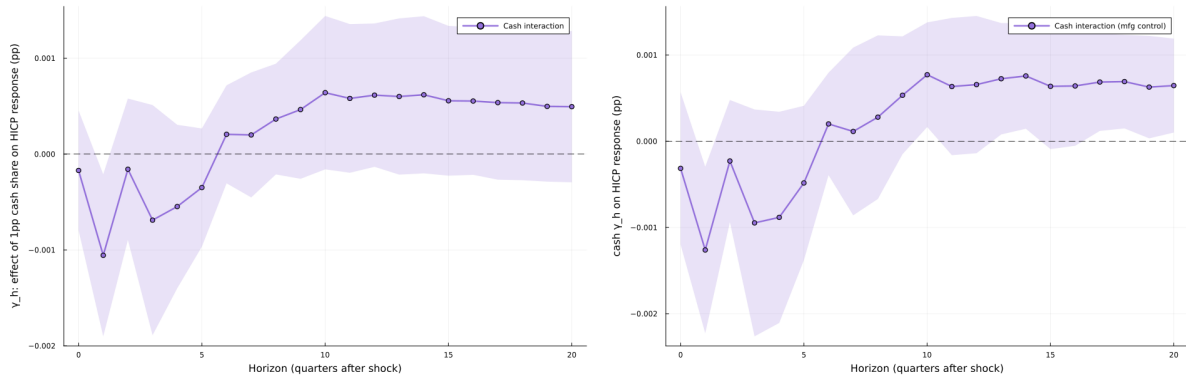


Figure 5: Horizon-varying cash interaction γ_h for the HICP response: baseline (left) and controlling for manufacturing share (right). 90% bands, standard errors clustered by country. Positive values indicate weaker price transmission in higher-cash countries.

6 Robustness

This section subjects the baseline output result, a positive trough cash interaction (quarters three to five) in the predicted direction but not statistically significant, to three checks, each entered separately: an industrial-structure control, a trade-openness control, and a leave-one-out exercise. It then re-examines the price response under a core price index. The effect's direction is robust to the industrial-structure control, under which the estimate sharpens, and to the leave-one-out exercise; adding trade openness leaves the cash coefficients little changed in magnitude but less precise, reflecting the moderate correlation between openness and cash.

6.1 *Industrial Structure*

Germany combines high cash usage with high manufacturing intensity, and manufacturing is rate-sensitive, so the baseline association might reflect industrial structure rather than payments. Re-estimating with the manufacturing share entered alongside cash and interacted with horizon, the cash coefficients are not attenuated (Table 3, right columns; Figure 6): at the trough they are, if anything, slightly larger (0.00142, 0.00172 and 0.00153 at quarters three to five) and more precisely estimated. Industrial structure therefore does not absorb the cash effect.

This check is evidence against one confounder, not the headline result: the primary estimate remains the baseline of Section 5, positive and correctly signed but not significant, and the thesis does not rest its claim on the greater precision of the manufacturing-controlled specification.

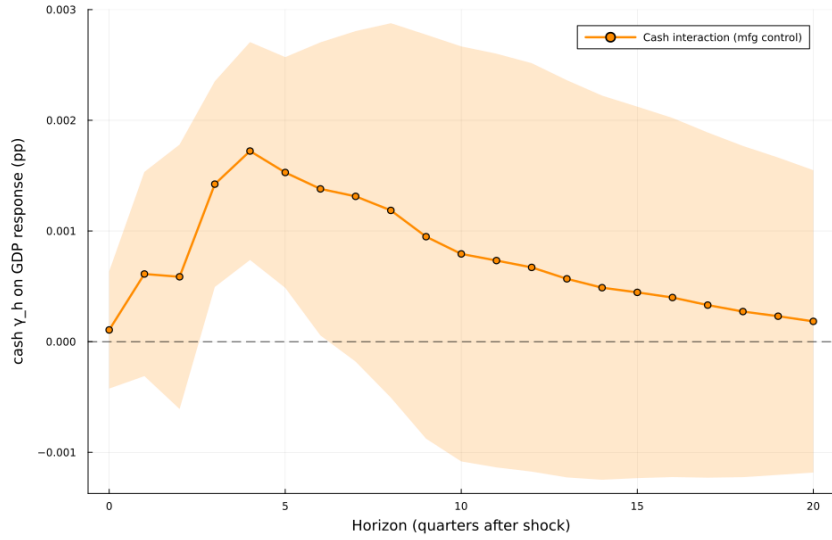


Figure 6: Cash interaction γ_h for the GDP response, controlling for manufacturing share interacted with horizon. 90% bands, standard errors clustered by country. Positive values indicate a shallower output contraction, that is weaker transmission, in higher-cash countries.

6.2 Trade Openness

A second characteristic that could overlap with cash is trade openness. Monetary policy transmits more weakly in more open economies, where the exchange-rate channel and demand leakage abroad dampen its effect on domestic output (Zhang, 2020); if openness correlated with cash usage, part of the apparent cash association could reflect it. Openness, constructed as in Section 3, is entered alongside cash and interacted with horizon, separately from the manufacturing control, since thirteen countries cannot support both at once.

Across the thirteen countries cash usage and trade openness are moderately and negatively correlated ($r \approx -0.38$): higher-cash economies tend to be somewhat less open, though the pattern is far from tight, with open high-cash economies such as Slovakia and Cyprus breaking it. When openness is added, the cash coefficients at the trough shrink, from 0.00111, 0.00130 and 0.00105 to 0.00072, 0.00102 and 0.00090 at quarters three to five, and remain statistically insignificant ($p = 0.44, 0.32$ and 0.39). With only thirteen countries, two slow-moving structural variables of this kind cannot be separately identified, the well-documented fragility of cross-country regressions (Levine & Renelt, 1992).

This secondary check does not overturn the output result. The cash estimates stay positive and of similar magnitude, so openness does not absorb the effect; and openness works, if anything, *against* the hypothesis. Because greater openness weakens transmission (Zhang,

2020) while higher-cash economies are on average slightly less open, openness predicts *stronger* transmission in exactly the higher-cash countries where the cash hypothesis predicts weaker transmission, so it cannot, on its own, generate the observed positive cash interaction.

6.3 *Influential Countries*

With only thirteen countries, the pooled estimate could in principle be driven by a single influential observation (Fox, 2020). To check this, the horizon-interaction regression is re-estimated thirteen times, dropping one country at a time. The average cash interaction across the trough window (quarters three to five) remains positive in every case, ranging from 0.00067 to 0.00162, so neither its sign nor its magnitude is the artefact of any single country; dropping Cyprus, the smallest economy in the sample, barely moves it, changing the average-effect p -value from 0.159 to 0.157. The estimate is, however, imprecise throughout: the average-effect p -value never falls below 0.065 (when Austria is dropped) and rises to 0.40 (when Finland is dropped). The direction of the effect is thus robust to leaving out any one country, even though it does not reach conventional significance in any of these re-estimations.

6.4 *Core Price Index*

The price results in Section 5 use the all-items HICP, which includes energy and unprocessed food, components that are volatile, driven largely by global commodity prices rather than domestic demand, and most exposed to the cost-push forces behind the price puzzle. A natural check is to repeat the analysis with the core HICP of Section 3, which excludes them and is transformed identically.

Of the thirteen baseline countries, twelve remain stable under the core index; only Austria fails, by the same mechanical, pre-specified criterion applied throughout (Section 4.3): swapping the price series pushes a root of its companion matrix to the unit circle, so its impulse response is no longer well defined. The all-items and core results are compared like-for-like on these same twelve countries, so any change reflects the price index alone, not the removal of Austria. On that common set the core index sharpens the profile in two directions. The hypothesis-consistent long-horizon effect strengthens: not significant at any individual long horizon under all-items, it is positive and significant under core from quarter ten onward, with bands above zero through quarter twenty and p -values around 0.03. As a single joint statement over quarters

ten to fourteen, the average core effect is positive and significant at the five-per-cent level ($\bar{\gamma} = 0.00064$, $t_{11} = 2.39$, $p = 0.036$), whereas the same twelve countries under all-items yield no significant average effect ($\bar{\gamma} = 0.00074$, $p = 0.154$). The early price puzzle, however, does not disappear; if anything it sharpens, turning significantly negative at quarters one, three, four and five under core against only quarter one under all-items, though with smaller coefficients (for example -0.00096 against -0.00131 at quarter one). It is therefore not simply a cost-push artefact of energy and food prices. On balance the core index strengthens the long-horizon, hypothesis-consistent evidence while leaving the short-horizon puzzle intact (Figure 7). The quarter-ten-to-fourteen window is the range at which, under the New-Keynesian price-lag logic of Section 7.1, a demand-driven price effect should surface once the faster cost-push response has passed.

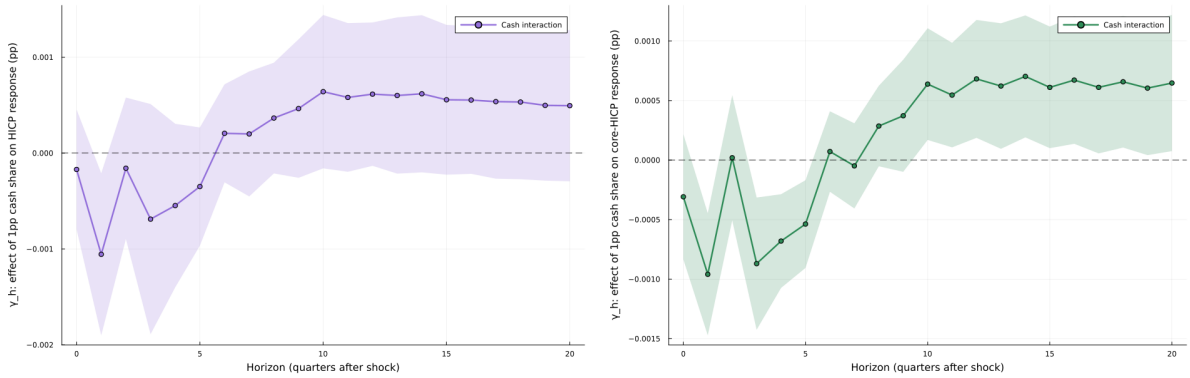


Figure 7: Horizon-varying cash interaction γ_h for the price response: all-items HICP across the thirteen-country baseline (left) and the core HICP, excluding energy and unprocessed food, across the twelve countries that remain stable under the core index (right). 90% bands, standard errors clustered by country. Positive values indicate weaker price transmission in higher-cash countries.

This pattern is what the interpretation in Section 7.1 would lead one to expect. If cash attenuates the demand response to a monetary shock, and prices respond to demand only sluggishly while the short-run price response is dominated by cost pass-through, then removing the volatile cost-driven components should reveal the demand-driven cash effect more clearly at longer horizons. The core-index results corroborate the all-items findings rather than replacing them, and strengthen the price-side evidence.

7 Discussion

7.1 Interpretation

The central finding is that higher cash usage is associated with a weaker output response to monetary policy at the horizons where transmission peaks, in the predicted direction and robust to dropping any single country. The effect on prices is weaker and more delayed. This pattern is broadly consistent with the theoretical framework, but the comparison deserves to be drawn carefully, because the literature does not imply that the output and price responses should behave identically.

For output, the bank-dependent channels emphasised by Mishkin (1995) (the bank-lending, balance-sheet and deposits channels) move spending through investment and bank-intermediated consumption, and these are the channels a high cash share most directly erodes. The result that higher-cash economies contract less is what this account predicts. It is worth asking whether the effect could instead have been zero or even reversed: if cash users are also more liquidity-constrained, the hand-to-mouth logic of Almgren et al. (2022) would push toward *stronger* responses in high-cash countries, working against the mechanism here. That the estimated effect is nonetheless positive suggests that the moderating effect of payment structure dominates that of liquidity constraints in this sample, though with thirteen countries the two cannot be cleanly separated.

For prices, the weaker and delayed response also fits the way price formation is usually modelled. In New-Keynesian accounts, prices respond to demand only sluggishly, through expectations and staggered price-setting, and much of the short-run price response to a monetary shock reflects cost pass-through rather than contemporaneous demand. If cash attenuates the demand response, the effect on prices should be both smaller and slower than the effect on output, which is what the long-horizon price profile, clearer under a core index, suggests, rather than absent. The question of why demand would adjust while prices adjust little is, in this reading, the standard one of nominal rigidity: weaker demand in higher-cash economies feeds into prices only gradually and is partly offset by common, area-wide cost-push forces captured by the commodity control. The output–price asymmetry is therefore consistent with cash weakening the real, bank-dependent channels more than the slower, expectations-driven process of price adjustment.

7.2 Limitations

Several qualifications bound the conclusions. Each horizon-specific coefficient is estimated from a cross-section of thirteen countries, so the trough effect should be read as suggestive rather than decisive. The result is best read as suggestive evidence whose credibility rests on the predicted sign, its concentration at the contiguous horizons coinciding with the pre-specified peak of the response, and its robustness to dropping any single country, rather than on statistical significance, which the baseline does not attain. This insignificance may reflect limited power rather than an absent effect: with only thirteen country responses, a cash share that varies over a compressed range, and the resulting imprecision in the cluster-robust standard errors, the data cannot sharply separate a weak or absent effect from a real one too imprecisely estimated to register. The correctly signed and economically meaningful point estimate is consistent with the former but does not establish it. Greater power, from more observations per country, from a local-projection design that retains the excluded countries, or from grouping countries, would help distinguish the two.

Controlling for manufacturing and, separately, trade openness (Section 6) does not eliminate every confounder. Trade openness in particular is moderately correlated with cash usage in this sample ($r \approx -0.38$), so the two are hard to separate across thirteen countries; the loss of precision when openness is added partly reflects this overlap, though it cannot be fully distinguished from a genuine openness effect, a boundary condition of small-sample cross-sectional designs rather than a refutation of the effect. Mortgage structure, the hand-to-mouth share, and the size of the informal economy could each contribute, and several of these are plausibly correlated with cash usage.

The informal economy is the most concerning, since cash-intensive economies tend to have larger informal sectors (Rogoff, 2017), and informality could weaken measured transmission independently of payment habits. This confounder also carries a measurement caveat: informal transactions are by definition unrecorded, so Eurostat’s real GDP captures only the formal economy. A monetary shock may affect informal activity differently from recorded activity, but that response cannot appear in the data, so the estimated moderation is bounded by the reach of official statistical tracking. It describes transmission to *measured* output rather than to all economic activity, and in the most cash-intensive economies the unmeasured margin is largest.

The hand-to-mouth share, as noted, works in the opposite direction (Almgren et al., 2022).

With thirteen countries, only a single such control can be entered alongside cash at a time before degrees of freedom are exhausted, which is why they are not all included; manufacturing and trade openness are each reported because industrial structure and external exposure are the most direct rate-sensitivity confounders. A fuller treatment would require either more countries or a framework that pools information across them.

The sample is also selected partly on VAR stability (Section 4.3), a property correlated with the kind of country involved, so the findings describe the thirteen survivors and need not extend to the rest. Because the common shock induces cross-sectional correlation in the residuals, clustering by country addresses dependence through time but not across countries within a quarter, and with so few clusters the cluster-robust errors are themselves imprecisely estimated; a wild cluster bootstrap would be the appropriate refinement. A more ambitious alternative would be to estimate all countries jointly in a hierarchical or Bayesian panel VAR, as in Mandler et al. (2016), which would pool information across countries, model the cross-sectional dependence directly, and let the cash moderator enter as a cross-country prior on the responses, at the cost of imposing more structure than the country-by-country design used here.

Two further methodological directions would address the design’s main constraints. First, the country-by-country impulse responses could be estimated by local projections with the high-frequency surprise as an external instrument (LP-IV), as in Almgren et al. (2022); because local projections estimate each horizon separately and do not require the companion matrix to be stable, this would avoid the stability-based exclusion of Section 4.3 and could retain the four dropped countries, enlarging the cross-section. Second, grouping countries, for example into high-cash and low-cash blocs or by region, and estimating responses at the group level would pool the limited time series within each group and sharpen the estimated responses, at the cost of the within-group heterogeneity the country-level design preserves.

Finally, the cash share is held fixed across the window. This abstracts from drift in payment habits: cash use declined substantially over the two decades of the sample, and not at the same rate in every country, so a time-invariant measure mismeasures the relevant share in the early part of the sample and, if the decline was faster where transmission was changing, could bias the estimated moderation.

The cross-sectional identification is nonetheless more defensible than the time-invariance

assumption alone suggests, because the regression exploits the *ranking* of countries by cash use, not the absolute level. Although cash use fell across the union, the ordering was persistent: the Netherlands and Finland remained among the lowest-cash economies and the southern and smaller member states among the highest throughout the sample, so a country's position in the cross-section (the variation the estimate actually uses) is more stable than its absolute share. Holding cash fixed also rules out, by construction, any reverse causation in which the strength of transmission influences cash use rather than the reverse, as could occur if economies where policy transmits weakly retain cash for reasons correlated with that weakness. A time-varying measure of cash use of comparable quality, were one available, would be needed to address both concerns; using an end-of-sample cross-section mitigates the second concern but not the first.

8 Conclusion

This thesis asked whether cash usage moderates the transmission of monetary policy in the euro area. Theory predicts that it should, since the bank-dependent channels of transmission run through a system that cash bypasses. The evidence bears out the two hypotheses unevenly. Hypothesis H1, that a higher cash share weakens the output response, finds qualified support: countries with higher cash usage show weaker GDP responses at the horizons where output troughs, in the predicted direction and robustly to dropping any single country. With only thirteen countries, however, the baseline effect is not statistically significant. Hypothesis H2, the analogous prediction for prices, is supported only weakly: the price response is weaker and more delayed, emerging clearly only at long horizons and only once the volatile components are removed in the core index. With only thirteen countries the output result is best regarded as suggestive evidence in favour of H1 rather than a definitive finding. The output–price asymmetry is exactly the pattern the New-Keynesian account of sluggish price adjustment would predict.

The analysis points to several extensions. Replacing quarterly GDP with monthly industrial production would roughly triple the observations per country and tighten the estimates. Adding the hand-to-mouth share of [Almgren et al. \(2022\)](#) as a control would address the most direct competing explanation. While this thesis infers that cash erodes the deposit-dependent channels from aggregate macroeconomic responses, future research could explicitly map these dynamics by adopting the Transmission Channel Analysis (TCA) framework developed by [Wegner, Lieb,](#)

Smeeke, and Wilms (2025). Applying their methodology to structural vector autoregressions with horizon interactions would allow a formal decomposition of the aggregate GDP impulse responses into channel-specific contributions, explicitly isolating the bank-lending or deposits channel from confounding structural characteristics and providing a direct test of whether the credit and interest-rate channels in particular are weaker where cash is more prevalent.

The findings also speak to the digital euro. If cash usage weakens output transmission, a central bank digital currency that draws cash transactions into the banking system would strengthen the reach of monetary policy, most of all in the high-cash economies of Austria, Italy and Spain. Such a currency would also draw currently cash-based, informal activity into the recorded financial system, shrinking the unmeasured margin that bounds the present estimates (Section 7.2) so that policy would reach a larger share of genuinely measured output (Rogoff, 2017). The size of any such gain would depend on the design of the currency, but the direction is consistent with the hypothesis, and it complements the efficiency and inclusion arguments already prominent in the policy debate.

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Appendix A Additional Results

Table 4 reports the country-level VAR estimates that feed the cross-country regression, and Table 5 gives the full horizon-by-horizon cash interaction summarised in Section 5.

Country	Cash (%)	Quarters	Lags	Peak GDP % (qtr)	Peak HICP % (qtr)
Netherlands	22	84	3	−0.077 (4)	−0.048 (10)
Finland	27	84	3	−0.113 (3)	−0.077 (10)
Luxembourg	37	84	3	−0.097 (3)	−0.075 (3)
Belgium	39	84	3	−0.061 (3)	−0.051 (6)
France	43	84	3	−0.048 (3)	−0.029 (3)
Cyprus	48	48	3	−0.028 (2)	−0.168 (4)
Germany	53	84	3	−0.064 (4)	−0.026 (5)
Greece	54	76	3	+0.019 (0)	−0.052 (3)
Portugal	54	84	3	−0.062 (3)	−0.038 (7)
Slovakia	57	44	3	−0.100 (12)	−0.103 (3)
Spain	57	84	3	−0.014 (3)	−0.059 (3)
Italy	61	84	3	−0.059 (3)	−0.056 (7)
Austria	62	84	3	−0.087 (3)	−0.040 (8)

Table 4: Per-country VAR summary for the thirteen baseline economies, ordered by cash share. “Peak” is the trough of the impulse response over the first twelve quarters, in per cent, with the horizon in parentheses. All systems are stable at three lags. Greece is the only country whose output response over this window is positive.

Horizon	GDP			HICP		
	γ_h	SE	p	γ_h	SE	p
q0	0.00006	0.00030	0.844	-0.00017	0.00035	0.635
q1	0.00049	0.00051	0.355	-0.00106	0.00047	0.046
q2	0.00038	0.00064	0.560	-0.00016	0.00041	0.709
q3	0.00111	0.00073	0.151	-0.00069	0.00067	0.326
q4	0.00130	0.00080	0.133	-0.00055	0.00048	0.275
q5	0.00105	0.00084	0.234	-0.00035	0.00035	0.333
q6	0.00089	0.00087	0.328	0.00021	0.00029	0.486
q7	0.00081	0.00088	0.380	0.00020	0.00037	0.595
q8	0.00067	0.00092	0.481	0.00037	0.00032	0.281
q9	0.00045	0.00095	0.640	0.00046	0.00041	0.274
q10	0.00034	0.00097	0.734	0.00064	0.00045	0.179
q11	0.00030	0.00096	0.760	0.00058	0.00043	0.207
q12	0.00027	0.00095	0.785	0.00061	0.00042	0.168
q13	0.00021	0.00093	0.829	0.00060	0.00046	0.213
q14	0.00016	0.00091	0.864	0.00062	0.00046	0.203
q15	0.00013	0.00088	0.882	0.00056	0.00044	0.229
q16	0.00010	0.00085	0.906	0.00055	0.00043	0.224
q17	0.00006	0.00081	0.944	0.00054	0.00045	0.257
q18	0.00002	0.00078	0.981	0.00053	0.00045	0.261
q19	-0.00001	0.00075	0.988	0.00050	0.00044	0.281
q20	-0.00004	0.00072	0.958	0.00049	0.00044	0.285

Table 5: Horizon-by-horizon cash interaction γ_h for the GDP and HICP responses, all twenty-one horizons, from the pooled regression of equation (4). Standard errors clustered by country; $G = 13$. The trough horizons q3–q5 are the focus of Section 5.

Appendix B Statement of Originality

By signing this statement, I hereby acknowledge the submitted thesis, entitled:

Monetary Policy Transmission in the Euro Area: Low-Cash versus High-Cash Economies

to be produced independently by me, without external help. Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is given.

By signing this statement, I explicitly declare that I am aware of the fraud sanctions as stated in the Education and Examination Regulations (EERs) of the SBE.

Place: Maastricht

Date: June 2026

Signature: Evlogjia Stefanidhi

Appendix C Declaration on the Use of Generative AI

In accordance with the SBE guidelines, the use of Generative AI tools in the preparation of this work is documented below.

During the preparation of this work, I used GenAI for the following purposes:

- Search engine: Claude (Claude Code) for locating relevant academic literature, verifying the bibliographic details of cited sources, and looking up technical syntax such as LaTeX.
- Ideation helper: Claude (Claude Code) for brainstorming and refining the framing of the empirical strategy and the robustness analysis.
- Text summarizer: Claude (Claude Code) for summarising academic papers and other source material.
- Explanation provider: Claude (Claude Code) for explaining econometric methods and concepts.
- Language assistant: Claude (Claude Code) for checking grammar and language and for revising prose for academic tone, structure, polishing sentences and clarity.
- Other: Claude (Claude Code) for assistance with writing and debugging the Julia analysis code and with preparing tables and figures.

After using any tool, I reviewed, quality-checked, and edited the content as needed and take full responsibility for the content of the thesis.